

Description of files in this folder

1. run_trim_galore.sh
 - a. Command used to run TrimGalore on the samples
 - b. FASTQs are available under NCBI BioProject ID PRJNA1328687
2. run_kallisto.sh
 - a. Command used to run kallisto after trimming
3. sample_info_transcriptomic_screen.xlsx
 - a. Used as metadata for the scripts below
4. pca.Rmd
 - a. Code used to produce PCA plots for the in vitro transcriptomic screen
 - b. Inputs:
 - i. The kallisto files which are available on GEO
 - ii. sample_info_transcriptomic_screen.xlsx (in this folder, see above)
 - c. Outputs – six PDFs in the pca_plots subfolder (used for multiple supplementary figures)
 - i. Note, PCA score plots may be reflected horizontally or vertically between runs because the signs of the principal components are arbitrary. These reflections do not change the interpretation of the PCA.
5. deseq2.Rmd
 - a. Code used to run DESeq2 for the in vitro transcriptomic screen
 - i. Inputs:
 1. The kallisto files which are available on GEO
 2. Reference transcriptome downloaded from Ensembl (release 110, Homo_sapiens.GRCh38.cdna.all.fa.gz)
 3. sample_info_transcriptomic_screen.xlsx (in this folder, see above)
 - ii. Outputs:
 1. DESeq2 outputs in the deseq2_outputs subfolder (used for supplementary tables)
 2. A supplementary figure showing the # DEGs per metabolite
 3. Correlation plots showing that vehicle controls that had the same solvent in the media, but were on different plates within a batch, could be grouped
6. lps_up_kegg_pathways_affected.Rmd
 - a. Inputs:
 - i. gprofiler outputs batch1-batch4.csv which are contained in the lps_up_gprofiler_kegg subfolder

1. These are gprofiler outputs based on the genes that LPS upregulated (see `deseq2_outputs` subfolder, above)
 2. These were used to create one of the supplementary tables
- b. Outputs:
 - i. PDFs `batch_1` through `batch_4` in that same subfolder (used for a main figure panel and a supplementary figure)
7. `num_degs_in_ad_pd_kegg_pathways.Rmd`
 - a. Inputs:
 - i. DESeq2 outputs (please see above)
 - ii. Lists of genes in Alzheimer's Disease / Parkinson's Disease KEGG pathways, downloaded from <https://rest.kegg.jp/link/hsa/path:hsa05010> and <https://rest.kegg.jp/link/hsa/path:hsa05012> on July 19, 2025
 1. These are present in this folder as `alzheimers_pathway_genes.txt` and `parkinsons_pathway_genes.txt`
 - b. Outputs:
 - i. A list of DEGs that overlap the AD/PD KEGG pathways, for each metabolite, within the `ad_pd_genes_affected` subfolder
 - ii. A figure panel showing the # AD/PD genes affected per metabolite
8. `chem_structure_svgs.py`
 - a. No inputs; outputs the four structure diagrams shown in main Fig. 2
 - i. These outputs are present in this folder as:
 1. `tmao.svg`
 2. `iaa.svg`
 3. `palmitate.svg`
 4. `gdca.svg`
 - b. rdkit must be installed to run this script
 - c. We used version 2025.03.4